

DESIGN AND IMPLEMENTATION OF A PARALLEL LINKAGE MECHANISM WITH SPRING ASSEMBLY FOR MAGNETIC FORCE COMPENSATION IN INSERTION DEVICES

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Abstract

This study presents the design and fabrication of a mechanical compensation system aimed at neutralizing the magnetic attraction forces inherent in insertion devices (IDs) used in synchrotron radiation facilities. In long IDs, such as those measuring 4 meters in length, multiple compensation modules, typically four, are required to maintain structural stability and magnetic field uniformity. In this work, a single compensation module was designed, fabricated, and installed on a test platform to verify the feasibility and mechanical performance of the proposed mechanism.

The system integrates a parallel linkage mechanism with a spring assembly consisting of twelve coil springs. The parallel linkage ensures synchronized and stable movement of the magnetic arrays with minimal structural deformation, while the spring assembly provides a counteracting force that balances the increasing magnetic attraction as the ID gap narrows. Although the mechanism was not installed on a working ID, test results demonstrate its effectiveness in reducing structural load and maintaining precise displacement control under simulated magnetic force conditions. This confirms the concept's viability and its potential for improving operational efficiency and safety in future ID applications.

INTRODUCTION

Insertion devices (IDs), such as undulators and wigglers, are essential components in synchrotron radiation facilities. These devices consist of opposing arrays of permanent magnets that generate highly periodic magnetic fields. By varying the gap between the magnet arrays, the spectral characteristics and intensity of the emitted synchrotron radiation can be finely controlled. However, a major challenge arises from the strong magnetic attraction forces that act between the two arrays, particularly when the gap is reduced. These forces increase nonlinearly as the gap narrows, posing significant mechanical challenges to the gap adjustment and support systems. Excessive magnetic forces can lead to structural deformation of the support beams, introduce positioning errors, and impose high loads on actuators, ultimately compromising beam quality and system reliability.

To mitigate these effects, various magnetic force compensation strategies have been implemented, including counterweights, hydraulic systems, magnetic force dynamic compensation, and spring-based mechanisms [1-4].

This study proposes a novel approach that integrates a parallel linkage mechanism with a spring assembly to provide passive compensation for the magnetic attraction forces in IDs. The parallel linkage ensures synchronized and vertically constrained motion of the magnetic arrays, thereby minimizing parasitic forces and improving motion precision. The spring assembly, composed of twelve coil springs, is designed to produce a counteracting force that closely follows the nonlinear increase of magnetic attraction as the gap decreases.

For long IDs (e.g., 4 meters in length), multiple compensation modules, typically four, are required to evenly support the structure and minimize deformation. In this work, a single compensation module was designed, fabricated, and installed on a test platform to evaluate the feasibility and performance of the proposed mechanism under simulated conditions. Although the system was not mounted on an actual ID, the experimental results confirm the design's effectiveness in compensating large magnetic forces and maintaining precise gap displacement. This work serves as a proof-of-concept and provides a solid foundation for future integration of such systems into operational insertion devices.

MAGNETIC FORCE IN INSERTION DEVICES

Insertion devices (IDs) utilize arrays of high-strength permanent magnets to generate periodic magnetic fields essential for producing synchrotron radiation. As the magnetic gap between the upper and lower magnet arrays decreases, the magnetic attraction force increases dramatically—often reaching several tons in large-scale systems. This substantial magnetic force poses a significant mechanical challenge, particularly in maintaining precise gap control and minimizing structural deformation.

In practice, the total magnetic attraction force increases linearly with the length of the ID, as each additional magnet segment contributes cumulatively to the overall force. For example, a 4-meter-long ID typically requires compensation for several tons of magnetic loading. The compensation system proposed in this study is designed such that each individual module can counteract up to 1.5 tons of magnetic force. Consequently, for a 4-meter-long ID, four compensation modules are strategically distributed along the device to ensure uniform load sharing and to minimize deflection of the beam supporting the magnet arrays.

The magnetic force F between permanent magnet arrays can be accurately approximated by an exponential function [5] of the form:

$$F(g) \approx F_0 \cdot e^{-k \cdot g} \quad (1)$$

where g is the gap, F_0 is the force at zero gap, and k is a configuration-dependent constant.

To facilitate analysis and practical implementation, the exponential relationship is approximated using six linear segments. Each segment is realized by combining different springs, with each configuration providing a distinct effective stiffness (K value) to fit a specific portion of the curve, as illustrated in Fig. 1. Except for the sixth segment where the magnetic force is relatively small, the force approximation error in the other segments is less than 10 kg by design.

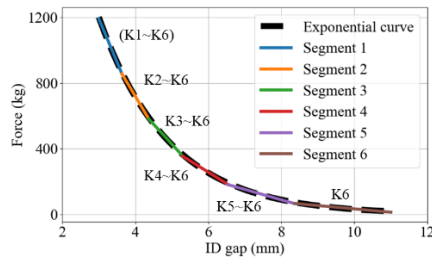


Figure 1: Magnetic attraction force versus ID gap, approximated by six linear segments, each realized through a specific spring set.

DESIGN REQUIREMENTS AND SYSTEM OVERVIEW

To effectively counteract the strong magnetic attraction forces present in insertion devices (IDs), the compensation mechanism must satisfy several critical design requirements:

- **Magnetic Force Compensation:** The system must generate a reliable and predictable counteracting force capable of balancing up to 1.5 tons of magnetic attraction per module. This compensating force must remain effective and stable across the full range of ID gap positions.
- **High Precision and Synchronized Motion:** The relative motion between the upper and lower magnet arrays must remain strictly parallel throughout the actuation range. Any angular deviation or misalignment can degrade the quality and uniformity of the magnetic field, adversely affecting synchrotron beam performance. The mechanism must ensure symmetrical, backlash-free, and repeatable motion.
- **Mechanical Stability and Structural Rigidity:** Due to the cyclic mechanical loading and long-term operation of IDs, the structure must maintain high stiffness and experience minimal deformation under load, ensuring consistent alignment and long-term mechanical integrity.

To meet these design criteria, the proposed compensation system incorporates a crossed four-bar linkage mechanism, also referred to as an X-shaped parallel linkage, as shown in Fig. 2. This kinematic configuration ensures synchronized, vertically constrained motion of the magnet arrays while minimizing parasitic lateral forces. The symmetrical motion paths offered by the X-shaped geometry

help maintain precise parallelism throughout operation and contribute to overall structural stability.

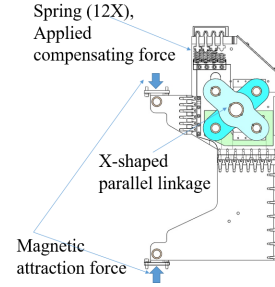


Figure 2: Overview of the magnetic force compensation module, featuring the X-shaped parallel linkage and spring assembly.

Integrated within the parallel linkage is a spring assembly consisting of twelve coil springs, arranged symmetrically and preloaded to deliver an upward compensating force. The spring force is designed to closely approximate the exponential profile of the magnetic attractive force as the gap narrows. This configuration forms a hybrid passive compensation system that reduces the load on active actuators, enhances system reliability, and maintains mechanical accuracy during dynamic operation.

Although a typical 4-meter-long ID would require four such compensation modules distributed along its length, this study focuses on the design and testing of a single module installed on a dedicated test platform. The goal is to verify its mechanical functionality and compensating performance prior to full-scale integration.

PARALLEL LINKAGE MECHANISM DESIGN AND ANSYS

To withstand the large vertical forces generated by magnetic attraction, the mechanism was designed with emphasis on high structural stiffness and mechanical integrity. A Finite Element Analysis (FEA) was conducted using SolidWorks Simulation to evaluate the system's response under full loading conditions. As shown in Fig. 3, only part of the simulation results is presented, confirming that the design satisfies the required strength and rigidity criteria, with deformation kept well within acceptable limits.

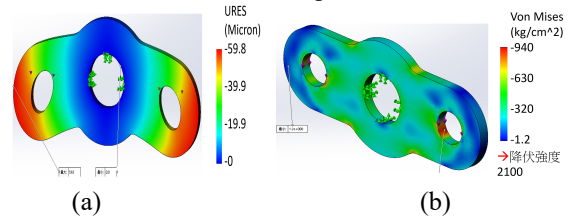


Figure 3. Finite Element Analysis results of the parallel linkage under a 1.5-tons vertical load. (a) Displacement distribution; (b) von Mises stress map.

To improve mechanical performance, reduce energy loss, and extend service life, several surface and motion optimization strategies were implemented:

- Low-friction, wear-resistant sliding pads were applied to contact surfaces to reduce friction while ensuring long-term durability.
- Self-lubricating bushings were used at all rotational joints to minimize maintenance needs and enable smooth, consistent motion.
- Hardened and precision-ground surfaces were applied to all sliding and guiding interfaces to enhance wear resistance, reduce frictional losses, and maintain dimensional stability under repetitive loading.

These design enhancements collectively improve system reliability, reduce backlash, and ensure consistent performance under cyclic operation conditions, as expected in synchrotron beamline environments.

EXPERIMENTAL VALIDATION

To evaluate the feasibility and performance of the proposed magnetic force compensation system, a complete parallel linkage mechanism integrated with a spring assembly was fabricated and mounted on a dedicated test platform.

The test platform, as shown in Fig. 4, replicates the mechanical conditions experienced in actual insertion devices (IDs). It is equipped with:

- A load cell for measuring the vertical compensating force generated by the entire mechanism (in kilograms),
- A touch sensor for recording the vertical displacement, corresponding to the simulated ID gap change (in millimeters).

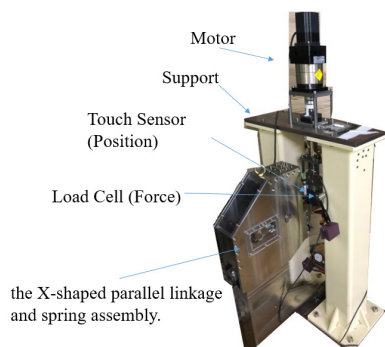


Figure 4. Photograph of the experimental test setup.

The relationship between applied force and gap displacement was recorded across a series of compression and decompression cycles. The results, illustrated in Fig. 5, indicate that the spring assembly provides a significant compensating force against the magnetic attraction. The force-gap curve exhibits high unidirectional repeatability, confirming the mechanical stability and consistency of the system.

The measurements show that the spring force closely matches the expected magnetic force profile across the tested gap range.

These experimental results validate the feasibility of using a parallel linkage mechanism with passive spring compensation for reliable and precise magnetic gap control in insertion devices.

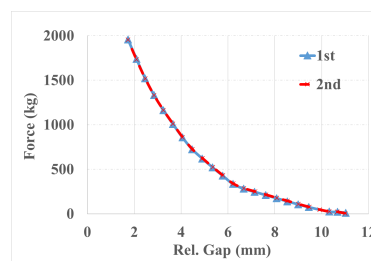


Figure 5. Test results showing the relationship between applied compensating force (kg) and ID gap displacement (mm). The curve demonstrates excellent repeatability and confirms that the spring force effectively counteracts the nonlinear magnetic attraction.

CONCLUSION

This study demonstrates the design and successful implementation of an X-shaped parallel linkage mechanism integrated with a spring assembly for magnetic force compensation in insertion devices (IDs). The proposed structure effectively enables strictly parallel motion of the magnetic arrays in response to gap changes, while minimizing the generation of undesired lateral or torsional forces. This inherent motion constraint and symmetry make the design particularly suitable for precision applications in synchrotron radiation systems.

The system has been validated experimentally and shown to effectively reduce magnetic loading with high repeatability and minimal deviation. Its simplicity and robustness make it a strong candidate for integration into actual insertion devices, especially those with long magnetic structures and high magnetic attraction forces.

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